AI-Based Detection of AI-Generated Images



TruthLens

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Project Overview:

A focused image-classification project, presented at the planning stage.

THE CORE QUESTION

Can an image be reliably classified as real or AI-generated?

The goal is to investigate and build a practical detection prototype—not to claim perfect detection.

- **Main objective**

  Develop an AI-based system to classify input images as real or AI-generated.

- **Planned outcome**

  A tested prototype with a simple interface for submitting an image and viewing a prediction.

- **Current scope**

  Still images only

CONCEPT STAGE

Literature Review:

Selected research will guide the methodology; the model choice remains open.

01

Problem & threat

How prior work defines AI-generated image detection and its challenges.

02

Methods explored

Compare approaches, data sources, evaluation settings, and reported limitations.

03

Relevance to TruthLens

Identify practical lessons that can guide dataset handling and model selection.

Detecting Deepfakes with Self-Blended Images (2022)

Video/Frame AUC / Ap:
- FF++ AUC: 99.64%
- Celeb-DF AUC: 93.18%
- DFD AUC:97.56%
- DFDC AUC:72.42%
- DFDCP AUC: 86.15%

Transcending Forgery Specificity with Latent Space Augmentation for Generalizable Deepfake Detection (2024)

- Avg DeepfakeBench AUC: 82.6%
- CDF-v1 AUC: 86.7%
- CDF-v2 AUC: 83.0% (Video AUC: 91.1%)
- DFDC AUC: 73.6% (Video AUC: 77.0%)

DF40: Toward Next-Generation Deepfake Detection (2024)

AUC / AP / EER:• Traditional detectors suffer severe drops on open-set forgeries (Xception avg AUC: 54.6%, SBI: 51.4%, RFM: 64.8%).
- CLIP baseline achieved avg open-set AUC: 69.2%

FakeRadar: Probing Forgery Outliers to Detect Unknown Deepfake Videos (2025)

Video-level AUC:
- Cross-dataset Avg AUC: 90.1%
- CDFv2 AUC: 91.7%
- DFDC AUC: 84.1%
- DFDCP AUC: 88.5%
- DFD AUC: 96.2%

Community Forensics: Using Thousands of Generators to Train Fake Image Detectors (United States University (2025)

The proposed model achieved 0.977 mean mAP and 0.966 mean accuracy across the evaluated benchmarks. On the high-resolution setting, it achieved 0.994 mean mAP and 0.986 mean accuracy.

Dataset & Planned Handling

The dataset will be inspected before committing to preprocessing or splitting decisions.

PLANNED SOURCE

CommunityForensics-Smal

Hugging Face dataset

Availability, labels, structure, and licensing details will be reviewed before use.

- **Inspect the dataset**

  Review class labels, file formats, metadata, and dataset organization.

- **Assess data quality**

  Check image integrity, duplicates, class balance, and potential sources of bias.

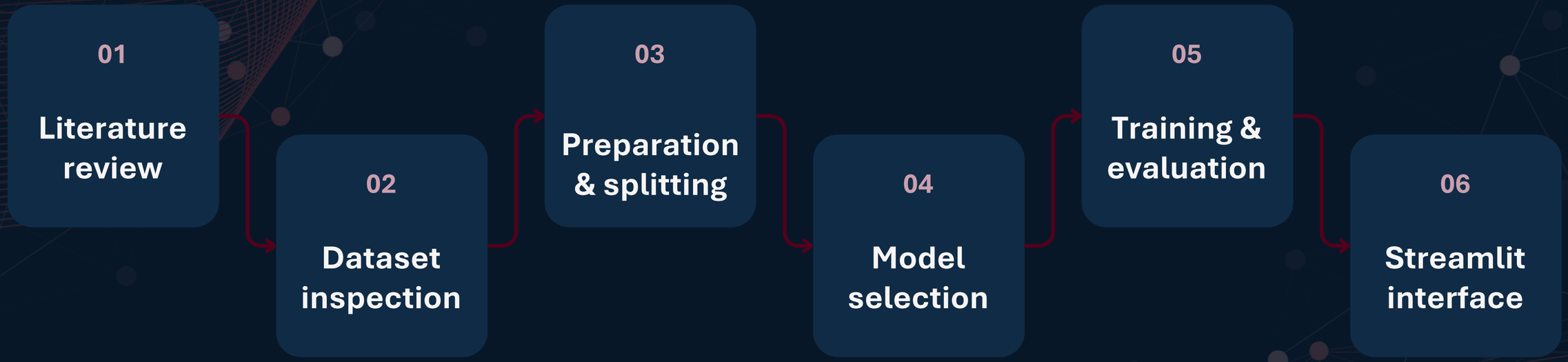
- **Plan preparation**

  Define suitable preprocessing and a leakage-aware train / validation / test split.

PLANNED WORK — DATA PREPROCESSING HAS NOT STARTED

Preliminary Pipeline

A working plan that may change after literature review and dataset inspection.



INTENDED OUTPUT A prototype that accepts a still image and returns a model prediction with appropriate caveats.

No model has been selected and no training or experiments have been conducted at this stage.

Preliminary Work Plan

The sequence will be refined with supervisor feedback and the findings from early investigation.

PHASE 01	**Research**	Review prior studies and summarize relevant methods.
PHASE 02	**Understand data**	Inspect dataset structure, labels, quality, and class distribution.
PHASE 03	**Design approach**	Decide preprocessing, splitting, and candidate-model comparison criteria.
PHASE 04	**Build & evaluate**	Train candidate model(s), evaluate performance, and analyze errors.
PHASE 05	**Prototype & report**	Build the Streamlit interface and document limitations and findings.

All phases are proposed; no completion dates or implementation results are claimed.

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- OwensLab, CommunityForensics-Small, Hugging Face Datasets: https://huggingface.co/datasets/OwensLab/CommunityForensics-Small